HOSTEL MANAGEMENT SYSTEM

SOFTWARE TEST PLAN AND TEST DOCUMENTATION

Member 2 — Testing Lead
Software Engineering Mini-Project — Part 1
Version 1.1 — Updated Requirements Baseline

Status: Finalized for Part 1 planning and test-case specification. Test execution results and defect statistics remain pending until an implemented system is available.

Document Control

Item	Value
Project	Hostel Management System (HMS)
Document Owner	Member 2 — Testing Lead
Version	1.1
Status	Finalized for Part 1 planning
Primary test documentation reference	ISO/IEC/IEEE 29119-3:2021
Related requirements reference	ISO/IEC/IEEE 29148:2018
Test basis	Member 1's updated SRS requirements, use cases, security objectives/requirements, and business rules

Revision History

Version	Date	Description
0.1	September 2026	Initial test-plan framework.
1.0	September 2026	Added requirement-based test cases and traceability.
1.1	September 2026	Updated test basis for revised SR-01, BR-03, UC-02, UC-09, FR-09, and FR-02.

1. Introduction

1.1 Purpose

This document defines the testing approach for the Hostel Management System (HMS), including test objectives, scope, strategy, levels and types, environment, test data, entry/exit criteria, defect management, risks, security validation, detailed test cases, and requirements traceability.

1.2 Scope

HMS covers room allocation, fee tracking, complaints, visitor logging, mess management, leave/outpass, and facility booking. Academic records and general accounting are excluded.

1.3 Actors

Actor	Responsibilities stated in SRS
Student	Complaints, fee payment, leave requests, mess menu viewing, facility booking; may view own profile but not edit it.
Warden	Leave approval, visitor logs, complaint handling.
Admin	Room allocation, student records, notices; creates and maintains student profiles.

1.4 References

- ISO/IEC/IEEE 29119-3:2021, Software and systems engineering — Software testing — Part 3: Test documentation.
- ISO/IEC/IEEE 29148:2018, Systems and software engineering — Life cycle processes — Requirements engineering.
- Hostel Management System SRS — Member 1's updated requirements baseline, September 2026.
- Project Software Architecture and Design Specification, when finalized.

2. Test Objectives

- Verify conformance to FR-01 through FR-09 and the applicable use cases.
- Evaluate NFR-01 response time, NFR-02 concurrency, NFR-03 availability, and NFR-04 usability.
- Verify SO-01 and SO-02, and security requirements SR-01 and SR-02.
- Verify business/validation rules BR-01 through BR-06.
- Detect, document, and support resolution of defects.
- Maintain traceability from requirements to test cases and test results.

3. Test Scope

3.1 In Scope

- Authentication and role-based authorization for Student, Warden, and Admin.
- Admin-managed student profiles and student read-only profile access.
- Room allocation and duplicate active-room prevention.
- Fee balance viewing, partial payments, and final Paid status.
- Complaint registration, status tracking, and complaint handling.
- Visitor entry/exit logging.
- Weekly mess menu publication and viewing.
- Leave/outpass requests and Warden approval/rejection.
- Admin notices and student visibility.
- Facility availability and booking, including slot-conflict prevention.
- Security, validation, integration, performance, availability, usability, and regression checks.

3.2 Out of Scope

- Academic records.
- General accounting.
- Features not specified in the SRS.
- Production operations beyond the evidence needed to assess the stated availability target.

4. Test Strategy

4.1 Test Levels

Level	Purpose
Unit	Verify individual units and validation/business logic.
Integration	Verify interactions among UI, REST API, authentication, and database.
System	Verify end-to-end HMS workflows against the SRS.
Acceptance	Check stakeholder-facing workflows against specified requirements.

4.2 Test Approach

- Positive, negative, boundary, and conflicting-state testing.
- Security testing for authentication, role permissions, data access, and error handling.
- Performance/load testing for NFR-01 and NFR-02.
- Availability measurement for NFR-03, excluding scheduled maintenance.
- Usability testing for NFR-04.
- Regression testing after fixes affecting existing behavior.

5. Testing Types

Type	Coverage
Functional	FR-01 to FR-09; UC-01 to UC-09.
Security	SO-01, SO-02, SR-01, SR-02.
Performance/load	NFR-01 and NFR-02.
Availability	NFR-03.
Usability	NFR-04.
Regression	Affected critical workflows after fixes.
Compatibility	Key workflows on the browsers supported by the final implementation.

5.1 Security Validation

ID	Validation	Expected Result
SEC-001	Attempt login with invalid credentials for	Authentication is denied.

	each role.	
SEC-002	Access a protected resource without authentication.	Request is rejected.
SEC-003	Attempt an Admin-only action as Student.	Action is rejected.
SEC-004	Attempt a Warden-only action as Student.	Action is rejected.
SEC-005	Attempt to access another user's restricted personal/payment data.	Unauthorized data is not disclosed.
SEC-006	Submit malformed or unexpected input.	Input is safely validated/rejected.
SEC-007	Use invalid/expired credentials or session token where applicable.	Protected access is rejected.
SEC-008	Trigger an invalid operation and inspect the error.	No unnecessary sensitive implementation details are disclosed.

6. Test Environment

Component	Planned environment
Frontend	React + TypeScript
Backend	Django + Django REST Framework
Database	PostgreSQL
API	REST
Styling	Tailwind CSS
Authentication	Implementation must satisfy SR-01; JWT was proposed, subject to final design.
Testing tools	Pytest and/or Django test framework, API tests, browser/system tests
Version control	Git + GitHub
Browsers	Chrome, Edge, Firefox, subject to final support statement

7. Test Data

- Valid and invalid accounts for Student, Warden, and Admin.
- Student profiles containing name, room, contact, and guardian information.
- Available/occupied rooms and students with/without active allocations.
- Fee accounts with outstanding balances; partial and full-settlement payment data.
- Complaints in Open, In Progress, and Resolved states.
- Visitor records with entry timestamps, exit timestamps, and active visits with null exit.
- Weekly mess menu data, leave requests, notices, and facility slots with both available and conflicting bookings.
- Synthetic data only; avoid real personal or payment data.

8. Entry and Exit Criteria

8.1 Entry Criteria

- Requirements are baselined and have unique IDs.
- A testable build and environment are available.
- Test accounts and representative data are prepared.
- Relevant APIs/database services are operational.
- Test cases have been reviewed.

8.2 Exit Criteria

- Planned critical/high-priority tests are executed or their blocked/not-applicable status is documented.
- Requirement-to-test traceability is reviewed.
- Critical/high defects are resolved or explicitly documented as accepted/open.
- Regression testing for significant fixes is completed.
- Execution evidence and a test summary are recorded.

9. Defect Management

Record defects in the agreed GitHub issue/defect-tracking mechanism. Each defect should be reproducible and linked to its test case.

Field	Description
Defect ID	Unique identifier.
Title	Short description.
Environment	Browser, build/version, and relevant configuration.
Preconditions / Steps	State and exact reproduction steps.
Expected / Actual	Required behavior versus observed behavior.
Severity / Priority	Impact and urgency.
Evidence	Screenshots, logs, request/response, or other evidence.
Status	Open, In Progress, Fixed, Retest, Closed, or Reopened.

Severity	Definition
Critical	Essential function failure or severe security/data-integrity impact.
High	Major function unusable or serious workflow impact.
Medium	Significant issue with a reasonable workaround.
Low	Minor issue with limited impact.

10. Test Risks and Mitigation

Risk	Impact	Mitigation
Late requirement changes	Tests may become outdated.	Review affected cases and RTM after changes.
Incomplete implementation	Tests cannot run.	Track readiness and prioritize critical workflows.
Insufficient test data	Scenarios may be missed.	Prepare representative synthetic data.
Unstable environment	Results may be unreliable.	Document and verify environment.
Late security findings	Rework may increase.	Test security early and during regression.
Limited time	Some tests may remain unexecuted.	Prioritize critical functionality and high-risk tests.
Regression from fixes	Previously working behavior may fail.	Run affected regression tests.

11. Test Deliverables

- Test Plan and Test Documentation.
- Detailed test cases and security test design.
- Requirements Traceability Matrix.
- Execution records and defect records.
- Final test summary and supporting evidence.

12. Roles and Responsibilities

Role	Responsibility
Member 2 — Testing Lead	Test plan, test cases, security testing, traceability, execution records, and test summary.
Member 1 — Requirements Lead	Maintain SRS and requirement IDs; communicate requirement changes.
Member 3 — Architecture Lead	Provide architecture information for integration/system testing.
Member 4 — Design Lead	Provide API/design/error-handling information for detailed tests.
Development team	Provide testable builds, fix defects, and support investigation.

13. Test Schedule

Phase	Activity	Dependency
1	Prepare test-plan framework and strategy.	None

2	Derive cases and traceability from SRS.	Updated Member 1 SRS — available
3	Refine API/integration cases.	Members 3 and 4 design details
4	Review test cases and data.	Testable implementation
5	Execute planned tests.	Implemented system
6	Retest fixes and perform regression.	Test execution and fixes
7	Compile final results and summary.	Execution complete

14. Detailed Test Cases

These 15 cases are the requirement-based Part 1 planning set. Execution status remains Not Executed until the system is implemented and tested.

TC-001 — Authenticate Admin, Warden, and Student

Field	Specification
Requirement ID(s)	UC-03, SR-01
Test Type	Functional / Security
Priority	High
Preconditions	Valid registered accounts exist for all three roles.
Test Data	Valid credentials for Admin, Warden, and Student.
Test Steps	For each role: open login; enter valid credentials; submit; verify role-appropriate access.
Expected Result	All three roles authenticate successfully. Passwords are stored using a secure hash, not plaintext.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	Verify password storage through an appropriate authorized test/configuration inspection; do not expose stored hashes in UI.

TC-002 — Enforce role-based access control

Field	Specification
Requirement ID(s)	SR-02, SO-02, UC-04, UC-05, UC-08
Test Type	Security / Authorization
Priority	Critical
Preconditions	Accounts for all roles exist.
Test Data	Student, Warden, Admin accounts.
Test Steps	As Student, attempt Admin room allocation and Warden leave approval. Then verify each role's permitted actions.
Expected Result	Unauthorized actions are rejected; permitted actions work only for the appropriate role; no unauthorized data is modified.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	

TC-003 — Allocate a room and prevent multiple active allocations

Field	Specification
Requirement ID(s)	FR-01, BR-01, UC-04
Test Type	Functional / Business Rule
Priority	Critical
Preconditions	A registered student has no active room; an available room exists.
Test Data	Student S1; available rooms R1 and R2.
Test Steps	As Admin, allocate R1 to S1; attempt to allocate R2 to S1 while R1 remains active.
Expected Result	First allocation succeeds. Second active allocation is rejected; S1 has at most one active room.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	

TC-004 — Maintain profile as Admin and enforce student read-only access

Field	Specification
Requirement ID(s)	FR-02

Test Type	Functional / Authorization
Priority	High
Preconditions	Admin and Student accounts exist; a student record exists.
Test Data	Name, room, contact, guardian information.
Test Steps	As Admin, create/update profile and save. As Student, view own record and attempt to edit it.
Expected Result	Admin can create/maintain the profile. Student can view their own record but cannot create or edit it; edit attempts are rejected.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	

TC-005 — Record partial payments and mark fee Paid only at zero balance

Field	Specification
Requirement ID(s)	FR-03, BR-03, SO-01
Test Type	Functional / Business Rule / Security
Priority	High
Preconditions	Student has an outstanding fee balance.
Test Data	Outstanding balance B; partial payment P where 0<P<B; remaining balance B-P.
Test Steps	As Student, view balance; make partial payment P; verify balance/status; pay remaining balance; verify status.
Expected Result	Partial payment is permitted and reduces the balance. Record is not marked Paid while balance remains. When balance reaches zero, status becomes Paid.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	

TC-006 — Enforce complaint status sequence

Field	Specification
Requirement ID(s)	FR-04, BR-04, UC-01
Test Type	Functional / Business Rule
Priority	High
Preconditions	Student can submit a complaint; complaint workflow is available.
Test Data	New complaint.
Test Steps	Submit complaint; verify Open; move to In Progress; move to Resolved; attempt to skip directly from Open to Resolved.
Expected Result	Valid transitions follow Open → In Progress → Resolved. Skipping a required state is rejected.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	

TC-007 — Record visitor entry and exit timestamps

Field	Specification
Requirement ID(s)	FR-05, BR-05, UC-05
Test Type	Functional / Validation
Priority	High
Preconditions	Warden is authenticated; student/room exists.
Test Data	Visitor details, entry timestamp, optional exit timestamp.
Test Steps	Create an active visitor log with entry and null exit; later add exit timestamp; attempt record without required entry timestamp.
Expected Result	Active visit may have null exit. Exit can be recorded later. A record missing required entry timestamp is rejected.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	

TC-008 — Publish and view weekly mess menu

Field	Specification
Requirement ID(s)	FR-06, UC-06
Test Type	Functional

Priority	Medium
Preconditions	Admin account exists; menu data is available.
Test Data	Weekly mess menu.
Test Steps	As Admin, publish/update menu. As Student, open mess menu.
Expected Result	Students can view the published weekly menu and see the latest published data.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	

TC-009 — Submit leave request and require Warden decision

Field	Specification
Requirement ID(s)	FR-07, BR-02, UC-02, UC-08
Test Type	Functional / Business Rule
Priority	Critical
Preconditions	Student and Warden accounts exist.
Test Data	Valid leave/outpass request.
Test Steps	As Student, submit request; verify pending state; confirm it is not Approved before Warden action; as Warden approve/reject; verify result.
Expected Result	Request remains pending until Warden action. Only authorized Warden action changes it to Approved or Rejected.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	

TC-010 — Broadcast Admin notice to students

Field	Specification
Requirement ID(s)	FR-08
Test Type	Functional / Authorization
Priority	Medium
Preconditions	Admin and Student accounts exist.
Test Data	Valid notice content.
Test Steps	As Admin, publish notice. As Student, open notices.
Expected Result	Notice is published by Admin and visible to the intended student audience.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	

TC-011 — Book available facility slot and reject conflict

Field	Specification
Requirement ID(s)	FR-09, BR-06, UC-09
Test Type	Functional / Business Rule
Priority	High
Preconditions	Student is authenticated; a facility slot is available.
Test Data	Facility F1; time slot T1.
Test Steps	View availability; book F1 for T1; attempt another booking for F1 at T1.
Expected Result	First booking succeeds. Conflicting booking is rejected and the original booking remains intact.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	

TC-012 — Verify standard request response time

Field	Specification
Requirement ID(s)	NFR-01
Test Type	Performance
Priority	High
Preconditions	Stable test environment and defined normal-load profile are available.
Test Data	Representative standard requests.

Test Steps	Run representative requests under normal load; measure response time using a documented method.
Expected Result	Response time satisfies the target of within 2 seconds under normal load.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	Before execution, define whether the threshold applies to each request, a percentile, or an aggregate.

TC-013 — Verify 500 concurrent student users

Field	Specification
Requirement ID(s)	NFR-02
Test Type	Performance / Load
Priority	High
Preconditions	Load-testing environment and representative data are available.
Test Data	At least 500 simulated concurrent student users.
Test Steps	Run representative student workflows with 500 concurrent users; monitor response, errors, and resources.
Expected Result	System supports at least 500 concurrent student users without unacceptable failures or incorrect behavior.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	Define duration, workload mix, acceptable error rate, and infrastructure before execution.

TC-014 — Verify 99% availability excluding scheduled maintenance

Field	Specification
Requirement ID(s)	NFR-03
Test Type	Availability
Priority	Medium
Preconditions	Monitoring is configured for an agreed observation period.
Test Data	Uptime/downtime records and scheduled-maintenance windows.
Test Steps	Collect service availability records; exclude scheduled maintenance; calculate availability.
Expected Result	Measured availability is at least 99% after excluding scheduled maintenance.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	Observation period and monitoring source must be agreed before execution.

TC-015 — Verify staff task-completion usability target

Field	Specification
Requirement ID(s)	NFR-04
Test Type	Usability
Priority	Medium
Preconditions	Representative staff participants and task scripts are available.
Test Data	Admin/Warden tasks based on specified workflows.
Test Steps	Give staff participants the tasks without prior training; record successful completions; calculate completion rate.
Expected Result	At least 90% of defined staff tasks are completed successfully without prior training.
Execution Status	Not Executed — implementation required
Defect ID	TBD during execution
Remarks	Define participant count, task list, success criteria, and calculation method before execution.

15. Requirements Traceability Matrix

The matrix links the supplied requirements, security objectives, and business rules to planned tests.

ID	Requirement summary	Test case(s)	Test type
FR-01	Admin allocates room to	TC-003	Functional

	registered student		
FR-02	Admin maintains profiles; Students may view but not edit own record	TC-004	Functional / Authorization
FR-03	Student views outstanding fees and records payments	TC-005	Functional
FR-04	Student registers complaint; status tracked to resolution	TC-006	Functional
FR-05	Warden logs visitor entry/exit against a student's room	TC-007	Functional
FR-06	Admin publishes weekly mess menu visible to students	TC-008	Functional
FR-07	Student submits leave/outpass; Warden approves/rejects	TC-009	Functional
FR-08	Admin broadcasts notices to students	TC-010	Functional
FR-09	Student views facility availability and books a facility	TC-011	Functional
NFR-01	Standard request within 2 seconds under normal load	TC-012	Performance
NFR-02	Support at least 500 concurrent student users	TC-013	Load
NFR-03	99% availability excluding scheduled maintenance	TC-014	Availability
NFR-04	Staff task-completion rate ≥90%, no prior training	TC-015	Usability
SO-01	Protect student personal/payment data from unauthorized access	TC-002, TC-005, SEC-005	Security
SO-02	Only authorized staff modify room allocation and fee records	TC-002, TC-003, TC-005	Authorization
SR-01	Hashed-password authentication for Admin, Warden, Student	TC-001, SEC-001, SEC-002	Security
SR-02	Role-based access control for Student/Warden/Admin	TC-002, SEC-003, SEC-004	Security
BR-01	No more than one active room per student	TC-003	Business rule
BR-02	Warden approval required before leave is Approved	TC-009	Business rule
BR-03	Partial payments allowed; Paid only at zero outstanding balance	TC-005	Business rule
BR-04	Complaint status Open → In Progress → Resolved	TC-006	Business rule
BR-05	Visitor entry timestamp required; exit may be null while active	TC-007	Validation
BR-06	Reject facility booking when slot is already booked	TC-011	Business rule

16. Test Execution and Results

PENDING IMPLEMENTATION/EXECUTION. Do not mark tests Pass or Fail until they have actually been run.

Field	Value
Execution status	Not yet executed
Build/environment	TBD
Execution date	TBD
Tester	TBD
Pass/Fail results	TBD
Linked defects	TBD

17. Defect Summary

PENDING TEST EXECUTION.

Metric	Value
Total defects	TBD

Critical	TBD
High	TBD
Medium	TBD
Low	TBD
Open	TBD
Closed	TBD
Reopened	TBD

18. Final SRS Alignment Notes

- SR-01 is updated to cover Admin, Warden, and Student logins; TC-001 and security validation reflect all three roles.
- BR-03 permits partial payments; TC-005 verifies that the balance decreases and the fee record becomes Paid only at zero balance.
- UC-02 and UC-08 are treated as sequential workflow steps, not a UML <<extend>> relationship.
- Facility terminology is aligned to booking; UC-09 is Book Facility and FR-09 covers availability and booking.
- FR-02 explicitly assigns profile creation/maintenance to Admin and read-only viewing to Students; TC-004 tests both permissions.

19. Remaining Work for Member 2

- Review any further changes to the SRS and update affected test cases/RTM.
- Refine API/integration/error-handling procedures after Members 3 and 4 finalize architecture and API design.
- Execute the 15 test cases once a testable implementation is available.
- Record actual results, dates, environment, evidence, and linked defects.
- Retest fixes and perform regression testing.
- Complete the final test summary and defect statistics.
- Export to PDF if the team submission requires PDF.

20. Conclusion

This version incorporates the updated requirements supplied by Member 1 and provides the Part 1 testing plan, security validation approach, 15 detailed test cases, and requirements traceability. Execution results and defect metrics remain pending until the implemented HMS can be tested.

21. References

- ISO/IEC/IEEE 29119-3:2021, Software and systems engineering — Software testing — Part 3: Test documentation.
- ISO/IEC/IEEE 29148:2018, Systems and software engineering — Life cycle processes — Requirements engineering.
- Hostel Management System SRS — Member 1's updated requirements baseline, September 2026.